

FOREX ENGINE PRO

SHADOW OS v2 Migration Program

SPRINT 0

Infrastructure Foundation

' SPRINT PASSED

Date: 2026-07-06
Version: 1.0
Git Commit: dc2a2791e97fb074b7df10c7ba51ae3a7d2fddf9
Database: heliumdb – PostgreSQL 16.10 (Railway)
Test Results: 19/19 PASS (8 smoke + 11 schema)
Gates: 7/7 PASSED
Data Lost: 0 rows

7

Gates
Passed

19

Tests
Passed

10

DB Tables
Created

8

Files
Archived

0

Data
Lost

1. Executive Summary

Sprint 0 establishes the non-negotiable infrastructure required before a single line of SHADOW OS v2 logic can be written. All six objectives were completed in a single session without any deployment, restart, or data-loss event.

The production bot (node telemetry/server.js) was not touched. All 29 rows of accumulated trading knowledge (events) and the live shadowm_trades record were preserved intact. The heliumdb PostgreSQL database now contains all 10 tables required by the SHADOW OS v2 architecture. A fully idempotent migration script and a 19-test validation suite verify that the foundation is solid.

Sacred Constraint (upheld throughout)

No deployment, restart, or migration step may ever destroy the accumulated trading knowledge of the system.

2. Objectives

#	Objective	Result
1	Archive all dead code with zero active dependencies broken	' COMPLETE
2	Create test framework using Node 24 built-in node:test	' COMPLETE
3	Write idempotent schema migration SQL for all 7 new tables	' COMPLETE
4	Build and validate the migration runner	' COMPLETE
5	Apply migration to production database with zero data loss	' COMPLETE
6	Produce Sprint 0 documentation	' COMPLETE

3. Completed Tasks

Task 1 — Dead Code Archive (GATE-0.A)

Identified 8 dead backup files with zero require() references in any active source file. Moved all 8 to archive/ preserving git rename-detection history.

Pre-archive safety check: grep -r require() across all 5 active source files returned 0 matches.

#	Original Path	Archive Path
1	dashboard.js	archive/dashboard.js
2	index_backup_v39_2.js	archive/index_backup_v39_2.js
3	index_backup_v39_3_before_v39_4.js	archive/index_backup_v39_3_before_v39_4.js
4	index_backup_v39_4_before_v39_4b.js	archive/index_backup_v39_4_before_v39_4b.js
5	index_original_safe.js	archive/index_original_safe.js
6	index_railway_mtf_v39_optimized.js	archive/index_railway_mtf_v39_optimized.js
7	telemetry/server_backup_pre_snowball_lab.js	archive/server_backup_pre_snowball_lab.js
8	telemetry/shadowlab_backup_pre_v40.js	archive/shadowlab_backup_pre_v40.js

Task 2 — Test Framework (GATE-0.B)

Created telemetry/tests/ with a four-tier structure using zero additional npm dependencies — Node 24's built-in node:test + node:assert/strict.

```
telemetry/tests/
  README.md           # Runner instructions, conventions, phase-gate policy
  unit/
    smoke.test.js     # 8 tests – environment, db-adapter, existing tables
    schema.test.js    # 11 tests – all 10 tables, indexes, constraints, idempotency
  integration/        # Placeholder – Sprint 1+
  stress/             # Placeholder – Sprint 3+
  mocks/             # Placeholder – Sprint 1+
```

Task 3 — Schema Migration SQL (GATE-0.C)

telemetry/migrations/001_shadow_os_v2_schema.sql (220 lines). All DDL uses CREATE TABLE IF NOT EXISTS and CREATE INDEX IF NOT EXISTS. All bootstrap inserts use ON CONFLICT DO NOTHING. Fully idempotent.

Task 4 — Migration Runner (GATE-0.D)

telemetry/migrations/run.js (179 lines). Validates DATABASE_URL, captures pre-migration row counts, executes SQL via psql -f --set ON_ERROR_STOP=1, verifies all 10 tables and 10 domain rows, compares post-migration row counts against baseline.

Task 5 — Migration Applied + Idempotency (GATE-0.D + GATE-0.E)

First run: created 7 new tables, 12 new indexes, 10 bootstrap rows. Second run: all IF NOT EXISTS / ON CONFLICT DO NOTHING applied, 0 errors, 0 changes.

Task 6 — Zero Active Code Modified (GATE-0.F)

```
git diff HEAD -- telemetry/server.js telemetry/shadowm.js telemetry/shadowlab.js
                  telemetry/db-adapter.js index.js
!' 0 lines changed
```

4. Files Changed

New Files

File	Purpose
archive/ (8 files)	Dead code with preserved git history
telemetry/migrations/001_shadow_os_v2_schema.sql	Idempotent DDL — 220 lines
telemetry/migrations/run.js	Migration runner (psql-based) — 179 lines
telemetry/tests/README.md	Test runner documentation
telemetry/tests/unit/smoke.test.js	8 smoke tests
telemetry/tests/unit/schema.test.js	11 schema validation tests
docs/reports/SPRINT_0_REPORT.md	This report (Markdown)
docs/reports/SPRINT_0_REPORT.pdf	This report (PDF)

Unchanged Active Production Files

File	Lines	Status
index.js (FROZEN — Railway entrypoint)	2360	' 0 lines changed
telemetry/server.js	2997	' 0 lines changed
telemetry/shadowm.js	718	' 0 lines changed
telemetry/shadowlab.js	1094	' 0 lines changed
telemetry/db-adapter.js	—	' 0 lines changed
telemetry/index.js	—	' 0 lines changed

5. Database Changes

Connection

Property	Value
Host	heliumdb (Railway)
Engine	PostgreSQL 16.10 on x86_64-pc-linux-gnu
Schema	public
Migration file	001_shadow_os_v2_schema.sql

New Tables Created

Table	PK	Key Features
runtime_domains	domain TEXT	Versioned JSONB state; 10 bootstrap rows
trade_intents	BIGSERIAL id	OANDA order idempotency; CHECK constraints
memory_entries	BIGSERIAL id	Namespace K/V; UNIQUE(ns,key); GIN tags
knowledge_artifacts	BIGSERIAL id	Versioned strategy snapshots; partial UNIQUE
event_idempotency	key TEXT	Deduplication; FK != events
consistency_log	BIGSERIAL id	Self-healing audit; CHECK on severity
system_snapshots	BIGSERIAL id	Full state capture at decision points

Existing Tables — Data Integrity

Table	Rows Before	Rows After	Result
events	29	29	' No data lost
shadowm_trades	1	1	' No data lost
shadowm_timeline	0	0	' No data lost

runtime_domains Bootstrap Rows (10)

Domain	Purpose
live	Daily trade counts, open positions, sequence counter
shadowA	Signal filter state (frozen=true initially)
shadowB	Signal confirmation state (frozen=true initially)
shadowC	KNN strategy selector state
shadowD	Condition weighting state
shadowM	Trade tracker state
exitLab	Exit strategy engine state
telemetry	Telemetry service state
scheduler	Cycle scheduler state
meta	System version, boot count, status

6. Test Results

Smoke Tests — 8/8 PASS

```
node --test --test-reporter=spec telemetry/tests/unit/smoke.test.js
```

#	Test	Result	ms
1	test framework is operational	' PASS	0.8
2	node:assert strict mode is working	' PASS	1.3
3	DATABASE_URL environment variable is set	' PASS	0.1
4	db-adapter module loads without error	' PASS	59.4
5	db-adapter connects and can run a basic query	' PASS	34.3
6	events table exists in the database	' PASS	3.1
7	shadowm_trades table exists in the database	' PASS	4.1
8	shadowm_timeline table exists in the database	' PASS	2.4

Schema Validation Tests — 11/11 PASS

```
node --test --test-reporter=spec telemetry/tests/unit/schema.test.js
```

#	Test	Result	ms
1	all 10 required tables exist	' PASS	36.0
2	runtime_domains has all 10 bootstrap rows	' PASS	4.1
3	runtime_domains columns are correct	' PASS	23.5

4	knowledge_artifacts has unique index on active artifacts	' PASS	7.6
5	trade_intents has partial index on PENDING status	' PASS	2.2
6	memory_entries has GIN index on tags	' PASS	2.5
7	consistency_log has correct severity CHECK constraint	' PASS	10.9
8	trade_intents has correct status CHECK constraint	' PASS	19.7
9	runtime_domains all rows have valid JSON values	' PASS	13.4
10	existing events and shadowm_trades data is intact	' PASS	4.4
11	migration is idempotent — running twice produces no errors	' PASS	4.0

Overall

Suite	Pass	Fail	Total
smoke.test.js	8	0	8
schema.test.js	11	0	11
TOTAL	19	0	19

7. Railway Validation

Check	Result
Production start command unchanged (node telemetry/server.js)	' Verified
railway.json not modified	' Verified
No new environment variables required	' Verified
No new npm packages added to production bundle	' Verified
index.js (Railway entrypoint) — 0 lines changed	' Verified
Migration runner is standalone, not part of server start	' Verified
Migration can be run: node telemetry/migrations/run.js	' Verified

No Railway deployment or restart was performed during Sprint 0. The migration runs against the production DB directly using DATABASE_URL and is completely independent of the bot lifecycle.

8. PostgreSQL Validation

Validation Step	Result
Connection established	'
psql available in environment (v16.10)	'
All 10 expected tables in information_schema.tables	'
All 10 runtime_domains rows with valid JSONB values	'
All CHECK constraints enforced (severity, intent_type, status)	'

All partial indexes present in pg_indexes	'
GIN index on memory_entries.tags present	'
Unique partial index on knowledge_artifacts (active)	'
Migration idempotent (second run: 0 errors, 0 new rows)	'
events row count: 29 != 29	' No data lost
shadowm_trades row count: 1 != 1	' No data lost
shadowm_timeline row count: 0 != 0	' No data lost

9. Production Validation

Check	Result	Notes
telemetry/server.js not modified	'	0 diff lines
telemetry/shadowm.js not modified	'	0 diff lines
telemetry/shadowlab.js not modified	'	0 diff lines
telemetry/db-adapter.js not modified	'	0 diff lines
index.js not modified	'	FROZEN — 0 diff lines
No new require() calls in production files	'	grep confirmed
Archived files have 0 active require() references	'	grep confirmed
Production bot behavior: zero change	'	No code paths altered
New DB tables are additive only	'	No existing tables dropped/altered

10. Risks

ID	Risk	Impact	Status
R-001	JS SQL splitter silently drops CREATE TABLE statements	High	& MATERIALIZED — Fixed
R-002	Migration destroys existing data	Critical	' Mitigated — row count guards
R-003	db.run() breaks on non-id PK tables	Medium	& MATERIALIZED — Documented
R-004	ANY(\$1) pg library array serialization bug	Medium	& MATERIALIZED — Fixed
R-005	New tables interfere with bot startup	High	' N/A — Sprint 0 is additive only
R-006	git mv blocked in Replit agent	Low	& MATERIALIZED — Used plain mv

11. Lessons Learned

L1 — Never use a JS regex-based SQL splitter for multi-statement DDL

L2 — db.run() auto-appends RETURNING id — dangerous for tables without an id column

telemetry/db-adapter.js appends RETURNING id to every INSERT lacking a RETURNING clause. Tables with non-BIGSERIAL primary keys (e.g. runtime_domains with PRIMARY KEY(domain)) throw 'column id does not exist'. Fix: use db.exec() for INSERTs into tables with non-id PKs.

L3 — ANY(\$1) with a JavaScript array returns zero rows in pg

Passing ["table1", "table2"] as a parameter to WHERE table_name = ANY(\$1) silently returns 0 rows due to pg's type resolution for information_schema.name columns. Fix: query all rows and filter in JS.

L4 — Node 24 test reporter flag is --test-reporter=spec, not --reporter=spec

--reporter=spec does not exist in Node 24's node --test. Process exits immediately with 'bad option'. Fix: always use --test-reporter=spec. Also note: exit code may be -1 even on all-pass — check output text.

L5 — psql is available in the Replit Nix environment

psql is at /nix/store/.../bin/psql at version 16.10 — no installation required. Migration scripts can reliably use psql as their execution engine.

12. Known Issues

#	Issue	Severity	Status
KI-00 1	db.run() fails on INSERT into tables with non-id PK (runtime_domains, event_idempotency)	Medium	Documented — use db.exec()
KI-00 2	node --test exits -1 even when all tests pass (Node 24 quirk with spec reporter)	Low	Check output text, not exit code
KI-00 3	git mv is blocked in Replit main agent	Low	Use plain mv — git detects rename
KI-00 4	git push from agent times out	Low	User must push from the Shell

13. Architecture Notes

SHADOW OS v2 Schema Overview

The SHADOW OS v2 schema is an event-sourced, domain-partitioned state store. The seven new tables form three logical tiers:

Tier 1 — State (runtime_domains)

Single-table key-value store where each row is an entire domain's mutable state. Version field enables optimistic locking. The 10 bootstrap rows pre-initialize all domain namespaces.

Tier 2 — Intent & Memory

- trade_intents: records every OANDA order attempt before it is sent, enabling idempotent retries and reconciliation after crashes.
- memory_entries: ephemeral and persistent K/V storage with namespace scoping, TTL, and GIN-indexed tags for fast multi-tag lookup.
- event_idempotency: prevents duplicate event processing by mapping idempotency keys to event IDs.

Tier 3 — Knowledge & Audit

knowledge_artifacts: versioned snapshots of trained strategy models (KNN weights, condition tables). A partial unique index ensures exactly one 'active' artifact per (domain, artifact) pair. Old versions are retained forever — knowledge is never deleted.

- consistency_log: append-only audit trail of self-detected data integrity issues, with severity levels and resolution tracking.
- system_snapshots: point-in-time captures of the full system state taken at key decision points.

Design Invariants

- Existing tables are never altered — only indexes are added with IF NOT EXISTS.
- All new tables are append-friendly — the bot can be restarted at any point without corrupt state.
- runtime_domains is the single source of truth for all mutable domain state.
- knowledge_artifacts uses a superseded_at pattern — knowledge is never deleted, enabling rollback and audit.

14. Migration Checklist

Step	Description	Status
MC-01	Confirm DATABASE_URL is set and points to PostgreSQL	' Done
MC-02	Verify psql is available in the execution environment	' Done
MC-03	Read and validate 001_shadow_os_v2_schema.sql for correctness	' Done
MC-04	Capture pre-migration row counts for all existing tables	' Done
MC-05	Run node telemetry/migrations/run.js (first pass)	' Done
MC-06	Verify all 10 tables exist in information_schema.tables	' Done
MC-07	Verify all 10 runtime_domains rows are present	' Done
MC-08	Verify post-migration row counts equal pre-migration counts	' Done
MC-09	Run node telemetry/migrations/run.js (second pass — idempotency)	' Done
MC-10	Confirm second pass produces 0 errors	' Done
MC-11	Run node --test smoke.test.js — 8/8 pass	' Done
MC-12	Run node --test schema.test.js — 11/11 pass	' Done
MC-13	git diff confirms 0 lines changed in active production files	' Done

15. Validation Checklist

Gate	Criterion	Pass Condition	Status
GATE-0.A	Dead code archived	8 files in archive/, 0 active require() refs	' PASS
GATE-0.B	Test framework operational	node --test runs; at least 1 test passes	' PASS
GATE-0.C	Migration SQL complete	All 7 new tables + indexes + bootstrap	' PASS
GATE-0.D	Migration runs clean	Runner exits 0; all 10 tables verified	' PASS
GATE-0.E	Migration is idempotent	Second run exits 0 with same output	' PASS
GATE-0.F	Zero active code modified	git diff on 5 active files = 0 lines	' PASS
GATE-0.G	All tests pass	19/19 pass (8 smoke + 11 schema)	' PASS

16. Sprint Status

' SPRINT 0 PASSED

Date: 2026-07-06 | Commit: dc2a2791e97fb074b7df10c7ba51ae3a7d2fbd9
Tests: 19/19 PASS | Data lost: 0 rows | Code changed: 0 lines
Gates: 7/7 PASSED

17. Readiness for Sprint 1

Sprint 1 Objective

Implement RuntimeDomainManager — the first SHADOW OS v2 domain manager. Responsible for reading/writing the runtime_domains table, providing atomic version-checked updates, and bootstrapping the domain state on first boot.

Sprint 1 Prerequisites

Prerequisite	Status
runtime_domains table with correct schema	' Present
10 bootstrap domain rows	' Present
consistency_log table for error recording	' Present
system_snapshots table for state capture	' Present
Test framework operational	' Operational
Migration runner validated and idempotent	' Validated
db-adapter.js available (no changes)	' Available
Zero active code changes required to unblock Sprint 1	' Confirmed

Sprint 1 Entry Criteria — ALL MET ' Sprint 1 may begin immediately.